

Virgo SLM V1.0

Building a 120M Parameter Transformer Language Model from Scratch

A Self-Learning Engineering Project

Tagline: *"Not Everybody Needs The Blueprint."*

Abstract

Virgo is an independent project built during my first year at IIT Dharwad to understand how modern language models are created. Rather than aiming to compete with frontier models, the goal was to learn every stage of development by building one from scratch. This report intentionally documents both the successes and the challenges.

Why I Started

I wanted to move beyond tutorials and research papers and experience the engineering process myself. This project became my learning notebook, where every bug, failed run, and improvement taught me something new. Suggested diagrams: personal roadmap, learning timeline.

Three Development Phases

Phase I – Language Learning: dataset collection, preprocessing, tokenizer, ~2B-token pretraining to create Virgo Base. Phase II – Chat Style Tuning: conversational datasets, assistant identity, dialogue quality, writing and coding improvements to create Virgo Chat. Phase III – Instruction & Response Optimization: a manually created instruction dataset of roughly 100 million tokens built over many hours. This phase teaches Virgo to better follow user instructions, produce structured responses, obey requested formats, and generalize across tasks, resulting in Virgo Instruct (IFT). Suggested diagram: three-stage roadmap.

Architecture

Describe tokenizer (45K Byte-Level BPE), decoder-only transformer (12 layers, 12 heads, hidden size 768), RoPE, FFN, residuals, and inference flow. Suggested diagrams: tokenizer workflow, transformer block, attention, RoPE, FFN.

Training Journey

Include hardware, checkpoints, mixed precision, training curves, token pipeline, engineering decisions, and a page titled 'Problems I Faced' describing CUDA OOM, preprocessing bugs, tokenizer issues, failed checkpoints, and debugging lessons.

Current Limitations

Virgo is currently limited in web search, image generation, advanced mathematical solving, high-level reasoning, file creation, long-context conversations, persistent memory, continuous multi-session conversations, and tool use. The current interaction model is prompt → response and does not maintain long-term memory between sessions.

Future Vision

Future work includes larger datasets, stronger reasoning, multilingual capability, better instruction following, longer context windows, multimodal understanding, tool use, memory, retrieval augmentation, improved evaluation, and eventually scaling beyond the Small Language Model category toward a larger language model while preserving efficiency.

Personal Message

Not Everybody Needs The Blueprint. This project was built independently as part of my learning journey. There were many failed experiments, long preprocessing jobs, training runs that did not work, and bugs that took hours to solve. Every setback became a lesson. If this report inspires another student to start building instead of waiting until they know everything, then Virgo has already achieved something meaningful. Curiosity, consistency, and patience matter more than having the perfect roadmap. Sometimes progress begins simply by taking the first step—even without a blueprint.

Recommended Diagrams (20–30)

Overall roadmap; learning timeline; dataset pipeline; preprocessing flow; tokenizer training; BPE example; tokenization example; transformer overview; decoder block; attention; RoPE; FFN; residual connection; training loop; optimizer flow; checkpoint system; inference pipeline; web application architecture; repository structure; dataset category pie chart; token distribution; training loss; learning rate; future roadmap; Virgo family tree.